make the second measurement without disturbing the state of the particle (the second collapse wouldn't change anything, in that case). But this is only possible, in general, if the two observables are compatible.

* **Problem 3.14**

(a) Prove the following commutator identities:

$$\left[\hat{A}+\hat{B},\hat{C}\right]=\left[\hat{A},\hat{C}\right]+\left[\hat{B},\hat{C}\right], \tag{3.64}$$

$$\left[\hat{A}\hat{B},\hat{C}\right]=\hat{A}\left[\hat{B},\hat{C}\right]+\left[\hat{A},\hat{C}\right]\hat{B}. \tag{3.65}$$

(b) Show that

$$\left[x^n,\hat{p}\right]=i\hbar n x^{n-1}.$$

(c) Show more generally that

$$\left[f(x),\hat{p}\right]=i\hbar\frac{df}{dx}, \tag{3.66}$$

for any function $f(x)$ that admits a Taylor series expansion.

(d) Show that for the simple harmonic oscillator

$$\left[\hat{H},\hat{a}_{\pm}\right]=\pm\hbar\omega\hat{a}_{\pm}. \tag{3.67}$$

Hint: Use Equation 2.54.

* **Problem 3.15** Prove the famous "(your name) uncertainty principle," relating the uncertainty in position ($A=x$) to the uncertainty in energy $\left(B=p^2/2m+V\right)$:

$$\sigma_x\sigma_H\geq\frac{\hbar}{2m}\left|\langle p\rangle\right|.$$

For stationary states this doesn't tell you much—why not?

Problem 3.16 Show that two noncommuting operators cannot have a complete set of common eigenfunctions. *Hint:* Show that *if* $\hat{P}$ and $\hat{Q}$ have a complete set of common eigenfunctions, then $\left[\hat{P},\hat{Q}\right]f=0$ for any function in Hilbert space.

3.5.2 The Minimum-Uncertainty Wave Packet

We have twice encountered wave functions that *hit* the position-momentum uncertainty limit $\left(\sigma_x\sigma_p=\hbar/2\right)$: the ground state of the harmonic oscillator (Problem 2.11) and the Gaussian wave packet for the free particle (Problem 2.21). This raises an interesting question: What is the *most general* minimum-uncertainty wave packet? Looking back at the proof of the uncertainty principle, we note that there were two points at which *in*equalities came into the argument: Equation 3.59 and Equation 3.60. Suppose we require that each of these be an *equality*, and see what this tells us about Ψ.

The Schwarz inequality becomes an equality when one function is a multiple of the other: $g(x)=cf(x)$, for some complex number c (see Problem A.5). Meanwhile, in Equation

3.60 I threw away the real part of z; equality results if $\mathrm{Re}(z) = 0$, which is to say, if $\mathrm{Re}\langle f|g\rangle = \mathrm{Re}(c\langle f|f\rangle) = 0$. Now, $\langle f|f\rangle$ is certainly real, so this means the constant c must be pure imaginary—let's call it ia. The necessary and sufficient condition for minimum uncertainty, then, is

$$g(x) = iaf(x), \quad \text{where } a \text{ is real.} \tag{3.68}$$

For the position-momentum uncertainty principle this criterion becomes:

$$\left(-i\hbar\frac{d}{dx} - \langle p\rangle\right)\Psi = ia(x - \langle x\rangle)\Psi, \tag{3.69}$$

which is a differential equation for Ψ as a function of x. Its general solution (see Problem 3.17) is

$$\Psi(x) = Ae^{-a(x-\langle x\rangle)^2/2\hbar}e^{i\langle p\rangle x/\hbar}. \tag{3.70}$$

Evidently the minimum-uncertainty wave packet is a *gaussian*—and, sure enough, the two examples we encountered earlier *were* gaussians.[22]

Problem 3.17 Solve Equation 3.69 for $\Psi(x)$. Note that $\langle x\rangle$ and $\langle p\rangle$ are *constants* (independent of x).

3.5.3 The Energy-Time Uncertainty Principle

The position-momentum uncertainty principle is often written in the form

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}; \tag{3.71}$$

Δx (the "uncertainty" in x) is loose notation (and sloppy language) for the standard deviation of the results of repeated measurements on identically prepared systems.[23] Equation 3.71 is often paired with the **energy-time uncertainty principle**,

$$\Delta t\,\Delta E \geq \frac{\hbar}{2}. \tag{3.72}$$

Indeed, in the context of special relativity the energy-time form might be thought of as a *consequence* of the position-momentum version, because x and t (or rather, ct) go together in the position-time four-vector, while p and E (or rather, E/c) go together in the energy-momentum four-vector. So in a relativistic theory Equation 3.72 would be a necessary concomitant to Equation 3.71. But we're not doing relativistic quantum mechanics. The Schrödinger equation is explicitly nonrelativistic: It treats t and x on a very unequal footing (as a differential equation it is *first*-order in t, but *second*-order in x), and Equation 3.72 is emphatically *not* implied by Equation 3.71. My purpose now is to *derive* the energy-time uncertainty principle, and in

[22] Note that it is only the dependence of Ψ on x that is at issue here—the "constants" A, a, $\langle x\rangle$, and $\langle p\rangle$ may all be functions of time, and for that matter Ψ may evolve away from the minimal form. All I'm asserting is that if, at some instant, the wave function is gaussian in x, then (at that instant) the uncertainty product is minimal.

[23] Many casual applications of the uncertainty principle are actually based (often inadvertently) on a completely different—and sometimes quite unjustified—measure of "uncertainty." See J. Hilgevoord, *Am. J. Phys.* **70**, 983 (2002).